

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	C03 – Precision (BL3, BL5)	Assessment:	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	Total Marks:	100
C03 Statement:	<i>Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.</i>		
Student Name:	SUMETHA SRI.S	Reg. No.:	192521097
Section / Batch:		Date:	29/07/2026

Code of Conduct

I SUMETHA SRI.S(192521097) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____Sumetha_____

Section / Topic	Focus	Q Nos	Marks
Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
GRAND TOTAL:		100 Marks	

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Annexure A – Interactive Tree Simulator

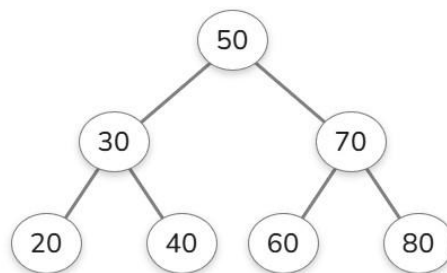
1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80 Perform:
 - a. Inorder Traversal
 - b. Search for 60
 - c. Delete 30
 - d. Display the final BST.
2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
 - a. Inorder
 - b. Preorder
 - c. Postorder
3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
5. Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Question 1



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Binary Search Tree

Input

Input an array to convert to a BST.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. e.g., 8, 3, 10, 1, 6, 14

Visualize

Balance the Tree

Generate Python Code

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80));
```

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INORDER TRAVERSAL

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Tree Traversals

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80));
```

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Tree Traversals

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

20 30 40 50 60 70 80

50

30

70

20

40

60

80

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SEARCH 60

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Binary Search

Input

Input a sorted array.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas, or generate default.

Target Value

60

Generate Default

Visualize

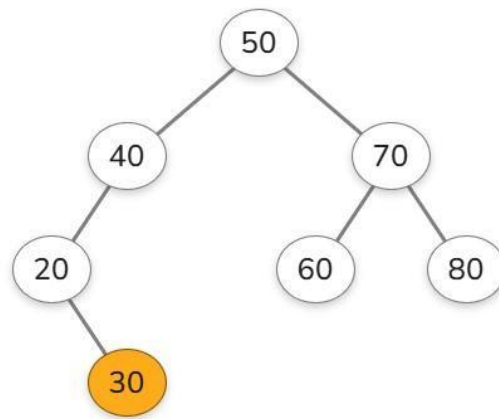
Start Simulation

20 30 40 50 60 70 80

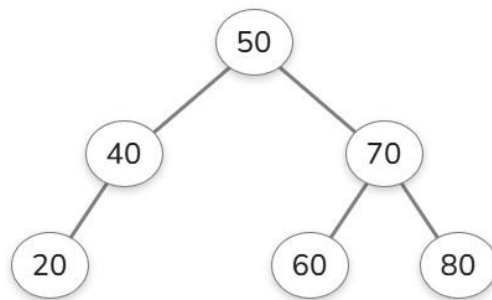
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DELETE 30



FINAL BST



Question 2

PRE ORDER

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Tree Traversals

?

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

PreInPost

Generate Default

Visualize

Start Simulation

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80));
```

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Tree Traversals

?

Input

Input is level-by-level order.

50, 30, 70, 20, 40, 60, 80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

PreInPost

Generate Default

Visualize

Start Simulation

```
graph TD; 50((50)) --> 30((30)); 50 --> 70((70)); 30 --> 20((20)); 30 --> 40((40)); 70 --> 60((60)); 70 --> 80((80));
```

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Tree Traversals

?

50302040706080

Input

Input is level-by-level order.

50,30,70,20,40,60,80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

PreInPost

Generate Default

Visualize

Start Simulation

Traversal result: [50, 30, 20, 40, 70, 60, 80]

OKAY

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INORDER

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Tree Traversals

?

Input

Input is level-by-level order.

50,30,70,20,40,60,80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

PreInPost

Generate Default

Visualize

Start Simulation



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Tree Traversals

20304050

Input

Input is level-by-level order.

50,30,70,20,40,60,80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

Generate Default

Visualize

Start Simulation

50

3070

20406080

Post

it

Traversal result: [20, 30, 40, 50, 60, 70, 80]

OKAY

POSTORDER

Tree Traversals

--	--	--	--	--	--	--	--

Input

Input is level-by-level order.

50,30,70,20,40,60,80

Traversal Order

Pre

In

Post

 Generate Default

► **Visualize**

▶ Start Simulation



Tree Traversals

?

20 40

Input

Input is level-by-level order.

50,30,70,20,40,60,80

Enter numbers separated by commas. Use 'null' for empty nodes.

Traversal Order

Pre

In

Post

 Generate Default

► **Visualize**

▶ **Start Simulation**

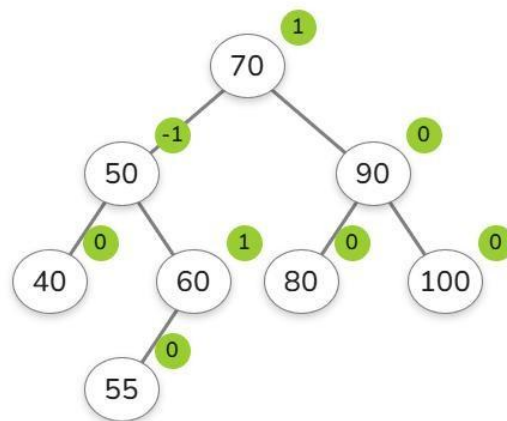


50

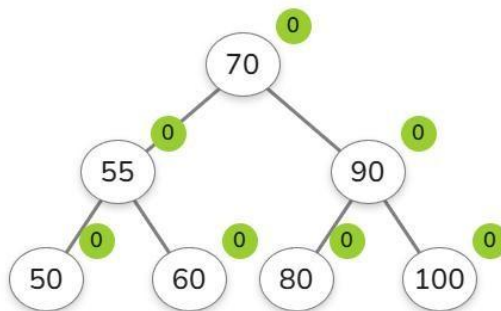
Traversal result: [20, 40, 30, 60, 80, 70, 50]

OKAY

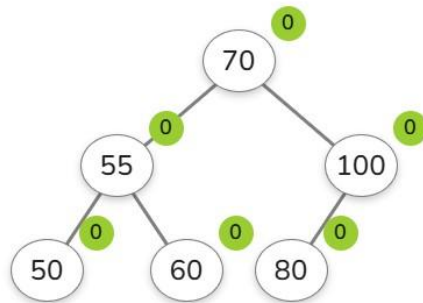
Question 3



Delete 40



Delete 90



Question 4

INSERT 45

45

INSERT 25

Enter a number: 25



INSERT

SEARCH

CLEAR



25

45

INSERT 65

Enter a number: 65



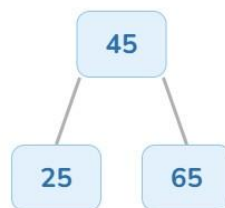
INSERT

SEARCH

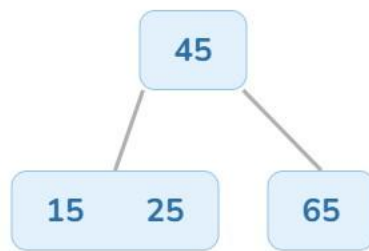
CLEAR



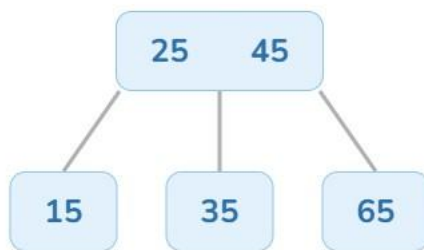
25 45 65



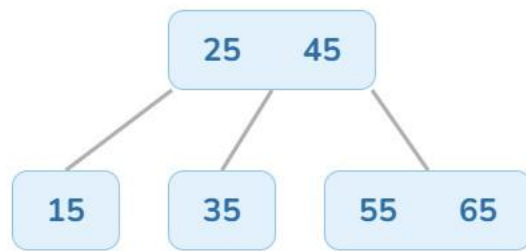
INSERT 15



INSERT 35



INSERT 55



INSERT 75

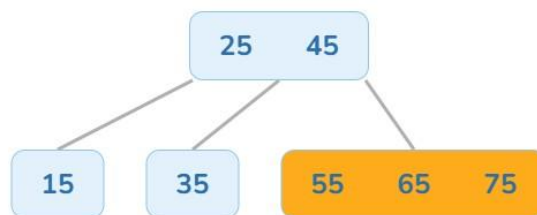
Enter a number: 75



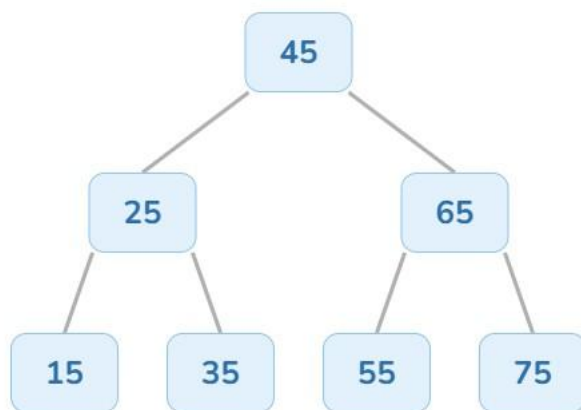
INSERT

SEARCH

CLEAR



FINAL B TREE



Question 5

